

Python Code Practice Set: 30 Questions

This practice set is based on the Python topics covered in the notebooks:

loops, lists, tuples, sets, dictionaries, file handling, JSON, OOP, list comprehensions, and lambda functions.

A. Loops

1. What is the difference between a `for` loop and a `while` loop in Python?
2. Write a `for` loop to print all numbers from `1` to `10`.
3. Write a `while` loop that prints numbers from `5` to `1` in reverse order.
4. Write a loop that prints all even numbers from `1` to `20`.
5. Write a loop that skips the number `5` and stops when it reaches `10`.

B. Lists

6. Create a list of five numbers and change the second element.
7. Use `append()` to add a new item to a list.
8. Use `extend()` to merge two lists.
9. Write code to insert a value at index `2` in a list.
10. Sort a list in ascending order and print both the original and sorted versions.

C. Tuples

11. What is a tuple, and how is it different from a list?
12. Create a tuple with three items and print the second item.
13. Write code to unpack a tuple into three variables.
14. Create a single-element tuple and print its type.
15. Write code to use `count()` and `index()` on a tuple.

D. Sets

16. Create a set from a list with duplicate values.

17. Add a new element to a set using `add()` .
18. Use `update()` to add multiple items to a set.
19. Write code to find the union of two sets and print the result.
20. Write code to find the intersection of two sets and print the result.

E. Dictionaries

21. What is a dictionary in Python?
22. Create a dictionary for a student with name, age, and grade.
23. Access a value using a key and print it.
24. Use `get()` to access a missing key with a default value.
25. Use `update()` to change one value and add one new key.

F. File Handling and JSON

26. Write code to create a text file and write a line into it.
27. Write code to read the contents of a text file using `with open()` .
28. Save a Python dictionary to a JSON file.
29. Read a JSON file and convert it back into a Python object.
30. Write code to append a new line to an existing file.

Challenge Tasks

- Write a small program that uses a list, a loop, and a condition together.
- Create a dictionary of student details and print all key-value pairs.
- Read and write a text file using `with open()` .
- Convert a Python dictionary to JSON and back.
- Create a class with one attribute and one method.